

NLP Hands on - NOTES Q&A	3
w1L1: Basic Language Processing Tasks, Tools and Algorithms	3
w1L2: What is NLP?	4
w1L3: Why is NLP hard?	7
w1L4: Empirical Laws	13
w1L5: Text Processing Basics	15
w2L1: Spelling Correction: Edit Distance	23
w2L7: Weighted Edit Distance, Other Variations	29
w2L8: Noisy Channel Model for Spelling Correction	35
w2L9: N-gram Language Models	41
w2L10 Evaluation of Language Models, Basic Smoothing	47
w3L12: w3L1: Language Modelling: Advanced Smoothing Models	53
w3L13: w3L2: Computational Morphology	63
w3L14: W3L3: Finite-state methods for morphology	69
w3L15: L3W4: Introduction to POS Tagging	74
w3L16: W3L5: Hidden Markov Models for POS Tagging	81
w4L17: W4L1: Viterbi Decoding for HMM, Parameter Learning	91
w4L18: W4L2: Baum Welch Algorithm	97
w4L19 W4L3: Maximum Entropy Models (Discriminative model)	103
w4L20 W4L4: Maximum Entropy Models (MEM model)	112
w4L21: W4L5: Conditional Random Fields	120
w5L22: w5L1: Syntax - Introduction	127
w5L23: w5L2: Syntax -Parsing I	131
w5L24: w5L3: Syntax -CKY, PCFGs	137
w5L25: w5L4: PCFGs - Inside-outside probabilities	142
w5L26: w5L5: Inside-outside probabilities	148
w6L26: W6L1: Dependency Grammars and Parsing - Introduction	153
w6L27: W6L2: Transition Based Parsing: Formulation	160

w6L28: W6L3: Transition Based Parsing: Learning	167
w6L29: W6L4: MST-based Dependency Parsing	177
w6L30: W6L5: MST-based Dependency Parsing: Learning	186
w7L31: W7L1: Distributional Semantics - Introduction	189
w7L32: W7L2: Distributional Models of Semantics	193
w7L33: W7L3: Distributional Semantics: Applications, Structured Models	198
w7L34: W7L4: Word Embeddings - Part I	203
w7L35: W7L5: Word Embeddings - Part II	209
w8L36: W8L1: Lexical Semantics (22-Sep-2021)	213
w8L37: W8L2: Lexical Semantics - WordNet	219
w8L38: W8L3: Word Sense Disambiguation - I	233
w8L39: W8L4: Word Sense Disambiguation - II	240
w8L40: W8L5: Novel Word Sense Detection (14min)	247
w9L41: W9L1: Topic Models: Introduction	251
w9L42: W9L2: Latent Dirichlet Allocation: Formulation	256
w9L43: W9L3: Gibbs Sampling for LDA, Applications	269
ToDo: from video copy handwritten notes	269
w9L44: W9L4: L44: LDA Variants and Applications - I	275
ToDo: handwritten notes A problem specified at 8min	275
w9L45: W9L5: LDA Variants and Applications - II	286
ToDo: handwritten notes	286
w10L46: w10L1: L47: Entity Linking - Part I	295
w10L47: W10L2: Entity Linking - Part II	303
w10L48: W10L3: Information Extraction - Introduction	311
w10L49: W10L4: Relation Extraction	317
w10L50: W10L5: Distant Supervision	322
w11L51: W11L1: Text Summarization - LexRank	331
w11L52: W11L2: Optimization Based Approaches for Summarization	341
w11L53: W11L3: Summarization: Evaluation (20min)	349

w11L54: W11L4: Text Classification - I	353
w11L55: W11L5: Text Classification - II	361
w12L56: Sentiment Analysis - Introduction	369
W12L2: Sentiment Analysis - Affective Lexicons	373
w12L58: L12W3: Learning Affective Lexicons	376
w12L59: W12L4 Computing with Affective Lexicons	381
Refer: Sentiment Tutorial by Christopher Potts	386
w12L60: W12L5: Aspect-based Sentiment Analysis	389
Exam Format:	393
Great Job!	393

NLP Hands on - NOTES Q&A

6-June-2021: NLP Course: <https://nptel.ac.in/courses/106/105/106105158/>

w1L1: Basic Language Processing Tasks, Tools and Algorithms

1. Basic Text Processing: Tokenization, Stemming, Spelling Correction
2. Language Modeling: N-grams, smoothing
3. Morphology, Parts of Speech Tagging
4. Syntax: PCFGs, Dependency Parsing
5. Lexical Semantics, Word Sense Disambiguation
6. Distributional Semantics, Word Embeddings
7. Topic Models

NLP Applications

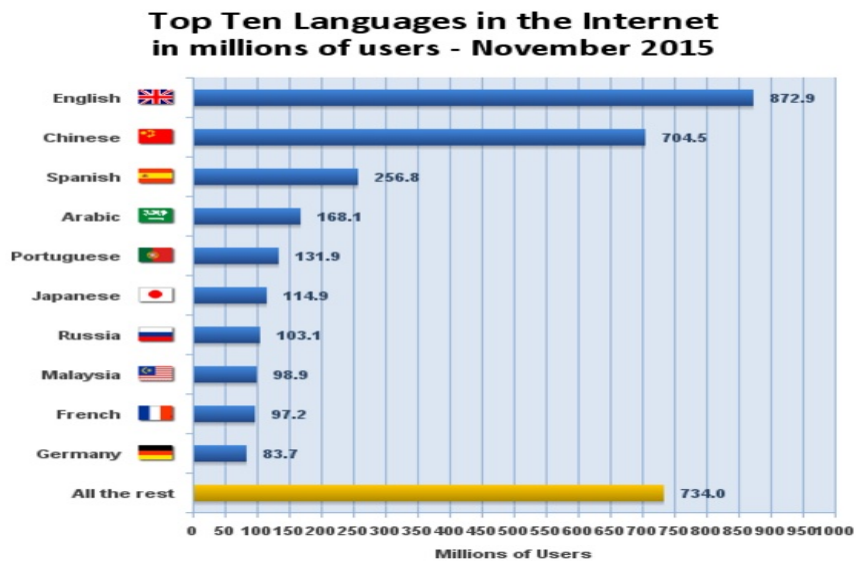
1. Entity Linking and Information Extraction
2. Text Summarization and Text Classification
3. Sentiment Analysis and Opinion Mining

Text is the largest repository of human knowledge

- news articles, web pages, scientific articles, patents, emails, government
- documents
- Tweets, Facebook posts, comments, Quora ...

Why study NLP?

You could not understand the majority of the world's data



w1L2: What is NLP?

Fundamental and Scientific Goal

Deep understanding of broad language (how humans understand language)

Om: Turing Goal

Engineering Goal

Design, implement, and test systems that process natural languages for practical applications

- Search, news reco

Goals can be very ambitious: Good quality translation

Google is awesome.

गूगल भयानक है।

googal bhayaanak hai.

Google is cool.

गूगल शांत है.

Gūgala śānta hai.

“Shant” meaning is not intended here

Well, even humans have made blunders

Pepsi Chinese blunder “*Come alive with the Pepsi Generation*”,

when translated into Chinese meant, “*Pepsi brings your relatives back from the dead.*”

KFC's Chinese blunder

KFC's slogan, "*Finger lickin' good*", when translated into Chinese meant "*We'll eat your fingers off.*"

Well, even humans ...

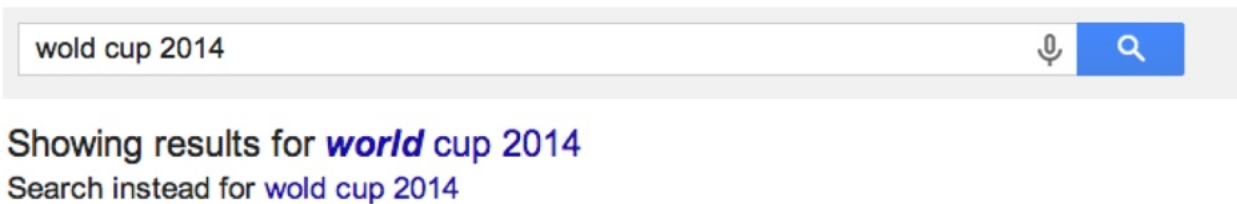


Goals can be very ambitious: Open Domain Chatbots

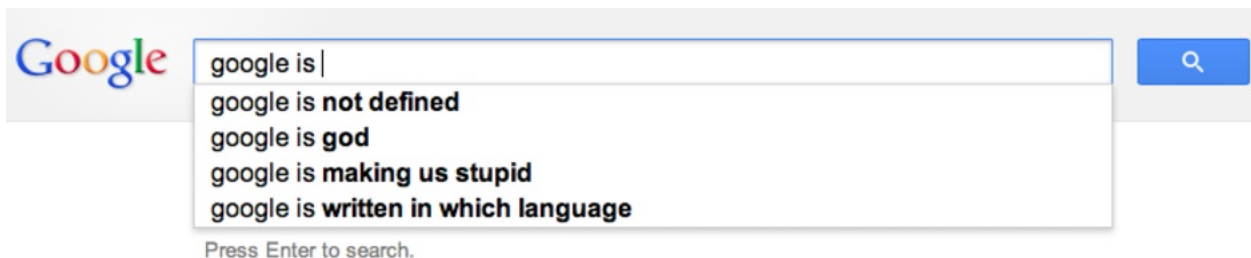
- *Taken down less than a day!*
- *Giving absurd and racist tweet*

And Goals Can be Practical: Auto Completion

- **Auto Suggest**



What you have queried or others have queried before



And Goals can be Practical: Information Extraction (from unstructured form)

Convert to structured form:

What are the entities and relations. Can you automatically build

New York Times Co. named Russell T. Lewis, 45, president and general manager of its flagship New York Times newspaper, responsible for all business-side activities. He was executive vice president and deputy general manager. He succeeds Lance R. Primis, who in September was named president and chief operating officer of the parent.

Person	Company	Post	State
Russell T. Lewis	New York Times newspaper	president and general manager	start
Russell T. Lewis	New York Times newspaper	executive vice president	end
Lance R. Primis	New York Times Co.	president and CEO	start

And Goals can be Practical: Domain-specific Chatbots

- Chatbot was used as TA
- 97% Accuracy
- Buy a product on ecommerce website, flight booking, banking system, chatbot can come with possible replies
- <http://www.news.gatech.edu/2016/05/09/artificial-intelligence-course-creates-ai-teaching-assistant>

Jill wasn't very good for the first few weeks after she started in January, often giving odd and irrelevant answers. Her responses were posted in a forum that wasn't visible to students.

"Initially her answers weren't good enough because she would get stuck on keywords," said Lalith Polepeddi, one of the graduate students who co-developed the virtual TA. "For example, a student asked about organizing a meet-up to go over video lessons with others, and Jill gave an answer referencing a textbook that could supplement the video lessons — same keywords — but different context. So we learned from mistakes like this one, and gradually made Jill smarter."

After some tinkering by the research team, Jill found her groove and soon was answering questions with 97 percent certainty. When she did, the human TAs would upload her responses to the students. By the end of March, Jill didn't need any assistance: She wrote the class directly if she was 97 percent positive her answer was correct.

And Goals can be Practical: **Sentiment Analysis**



Other Goals

- Spam detection
- Machine Translation services on the Web
- Text Summarization
- ...

Natural Language Technology not yet perfect

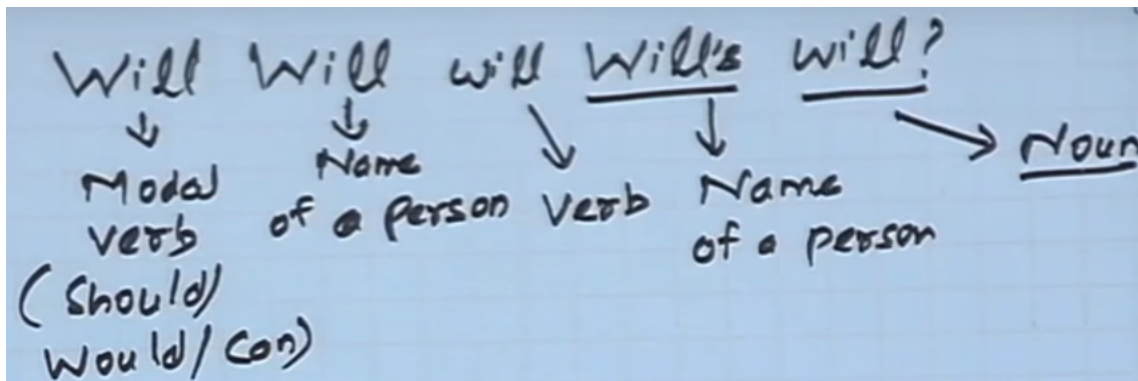
But still good enough for several useful applications

Help Society !

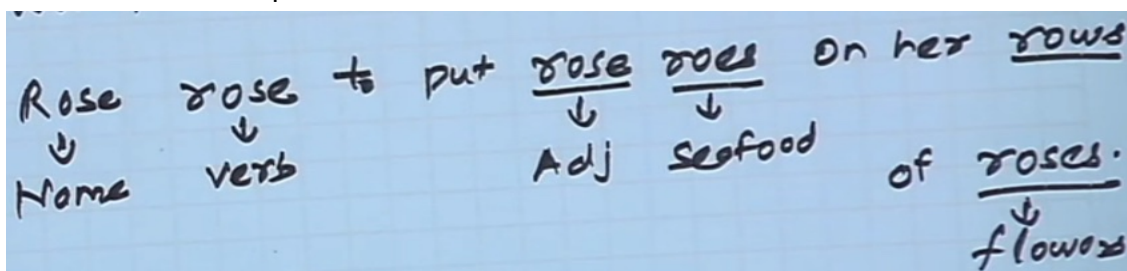
w1L3: Why is NLP hard?

1. Lexical Ambiguity (same word used differently)

- Will Will will Will's will?
- Rose rose to put rose roes on her rows of roses.
- Buffalo buffalo Buffalo buffalo buffalo buffalo Buffalo buffalo.
- Buffaloes from Buffalo, NY, whom buffaloes from Buffalo bully, bully buffaloes from Buffalo.
- Will Will will Will's will?



- Rose rose to put rose roes on her rows of roses.



- Buffalo buffalo Buffalo buffalo buffalo Buffalo buffalo. (8 times)

Buffalo → City in US
Buffalo → animal

buffalo → verb
(bully)

Buffalo - city, A - Animals, V - verbs

✓ Buffalo buffalo Buffalo buffalo buffalo
C A C A C A

buffalo Buffalo buffalo
V C A

- Buffaloes from Buffalo, NY, whom buffaloes from Buffalo bully, bully buffaloes from Buffalo.

2. Language ambiguity: Structural (different meaning)

- The man saw the boy with the binoculars.
- Who has binoculars: man or boy?

The man saw the boy with the binoculars.

- Flying planes can be dangerous.

Flying planes can be dangerous.

- Is flying dangerous or flying planes is dangerous?
- Hole found in the room wall; police are looking into it.
 - Is police looking into the hole or in the case?

3. Language imprecision and vagueness

- It is very warm here.
 - India, Europe, whether last month - relative
- Q: Did your mother call your aunt last night?
- A: I'm sure she must have. (**means = I don't know**)

But that's the fun part of it

Why is the teacher wearing sun-glasses?

...

Because the class is so bright. (student or sunlight)

Ambiguities

News Headlines

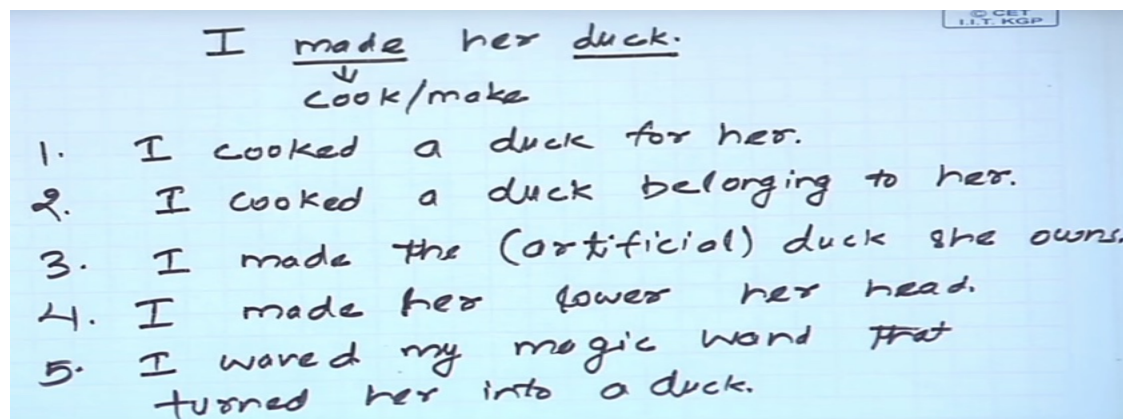
- Hospitals Are Sued by 7 Foot Doctors
 - there are 7 different 'foot' doctors
- Stolen Painting Found by Tree
 - Paining was not found by Tree but was near tree
- Teacher Strikes Idle Kids
 - Semi colon was missing.

Ambiguity is pervasive

Find at least 5 meanings of this sentence:

I made her duck.

Made = cook



Ambiguity is pervasive

Find at least 5 meanings of this sentence:

- I I made her duck
- I cooked duck for her
- I cooked duck belonging to her
- I created the (artificial) duck, she owns
- I caused her to quickly lower her head or body
- I waved my magic wand and turned her into a duck

What is in the language:

Syntactic Category

'Duck' can be a **noun** (n) or **verb** (v)

'her' can be a **possessive** ('of her') or **dative** ('for her') pronoun

Word Meaning

'make' can mean 'create' or 'cook'

Grammar

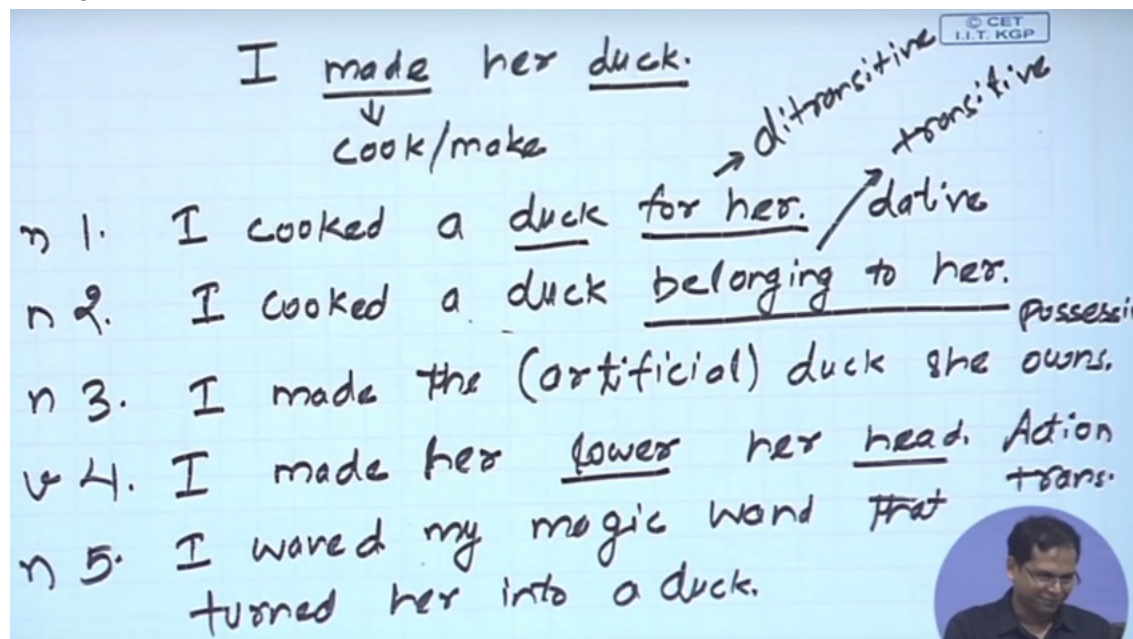
make can be

- Transitive: (verb with a noun direct object)
- Ditransitive: (verb has 2 noun objects)
- Action-transitive: (verb has a direct object + verb)

Phonetics

I'm eight or duck

I'm aid her duck



Ambiguity is Explosive

I saw the man with the telescope. 2 parses

I saw the man with the telescope. 2 parses

I saw the man on the hill with the telescope. 5 parses

I saw the man on the hill in Texas with the telescope. 14 parses

I saw the man on the hill in Texas with the telescope at noon. 42 parses

I saw the man on the hill in Texas with the telescope at noon on Monday.

132 parses

Catalan number

Refer: Catalan Number

Why is Language Ambiguous?

The goal in the production and comprehension of natural language is *efficient* communication.

Allowing resolvable ambiguity

- permits shorter linguistic expressions
- avoids language being overly complex

Language relies on people's ability to **use their knowledge** and **inference abilities** to properly resolve ambiguities

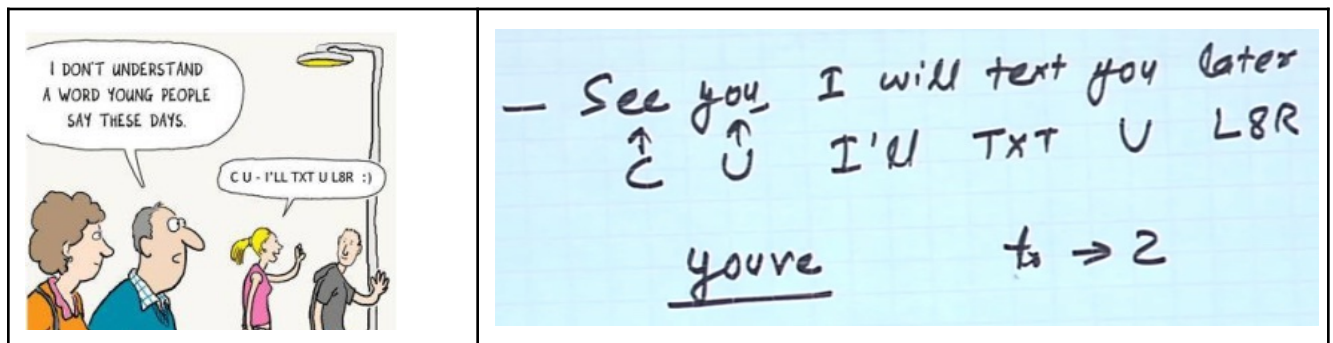
Natural Languages vs. Computer Languages

Ambiguity is the primary difference between natural and computer languages.

Formal programming languages are designed to be unambiguous

- Formal programming languages can be defined by a grammar that produces a **unique parse** for each sentence in the language.

Programming languages are also designed for **efficient (deterministic) parsing**.



Non-standard English (*in social media SMS*)

Great job @justinbieber! Were SOO PROUD of what youve accomplished! U taught us 2 #neversaynever & you yourself should never give up either

Segmentation Issues

the New York-New Haven Railroad

the [New] [York-New] [Haven] [Railroad]

the [**New York**]-[**New Haven**] [Railroad] - *correct (find what is correct segmentation)*

Idioms

- dark horse (*can't take individual words as black horse*)
- Ball in your court (*the matter is at your hand, it is your turn*)
- Burn the midnight oil (*doing hard work*)

Neologisms (new words, new meaning)

- Unfriend (*on facebook, has become work*)
- retweet
- Google/Skype/photoshop (*googling*)

You don't have close vocabulary

New Senses of a word

That's sick dude! (sick = unhealthy)

Giants ... *multinationals, conglomerates, manufacturers*

Tricky Entity Names

Where is A Bug's Life playing ... ('A Bug's Life' is single entity)

Let It Be was recorded ... (Let it be - single entity)

Tools Required

- Knowledge about language
- Knowledge about the world
- A way to combine knowledge resources

How is it generally done?

Probabilistic models built from language data

- $P(\text{"maison"} \mid \text{"house"})$ is high
- $P(\text{I saw a van}) > P(\text{eyes awe of an})$

Extracting rough text features does half the job.

w1L4: Empirical Laws

Function Words vs. Content Words

Function words have little lexical meaning but serve as important elements to the structure of sentences.

Function (a, the)

Example

Find Content word replaced by garbage word (not in language)

Find Function word replaced by garbage word (not in language)

The winfy prunkilmonger from the glidgement mominkled and brangified all his leversers vederously. (replaced instructor with 'winfy prunkilmonger')

- **Can understand - structure only structure**

Glop angry investigator larm blonk government harassed gerfritz infuriated sutbor pumrog listeners thoroughly. (**The replaced by Glop**)

- **Can understand - structure and topic from second sentence**

Function words are closed-class words

prepositions, pronouns, auxiliary verbs, conjunctions, grammatical articles, particles etc.

- **Will not have newer words coming in**

Most Common Words in Tom Sawyer (most frequently used words)

Word	Freq.	Use
the	3332	determiner (article)
and	2972	conjunction
a	1775	determiner
to	1725	preposition, verbal infinitive marker
of	1440	preposition
was	1161	auxiliary verb
it	1027	(personal/expletive) pronoun
in	906	preposition
that	877	complementizer, demonstrative
he	877	(personal) pronoun
I	783	(personal) pronoun
his	772	(possessive) pronoun
you	686	(personal) pronoun
Tom	679	proper noun
with	642	preposition

The list is dominated by the little words of English, having important grammatical roles.

What is special about which category you see

- Most of the words are function words (also known as **Stop words**)

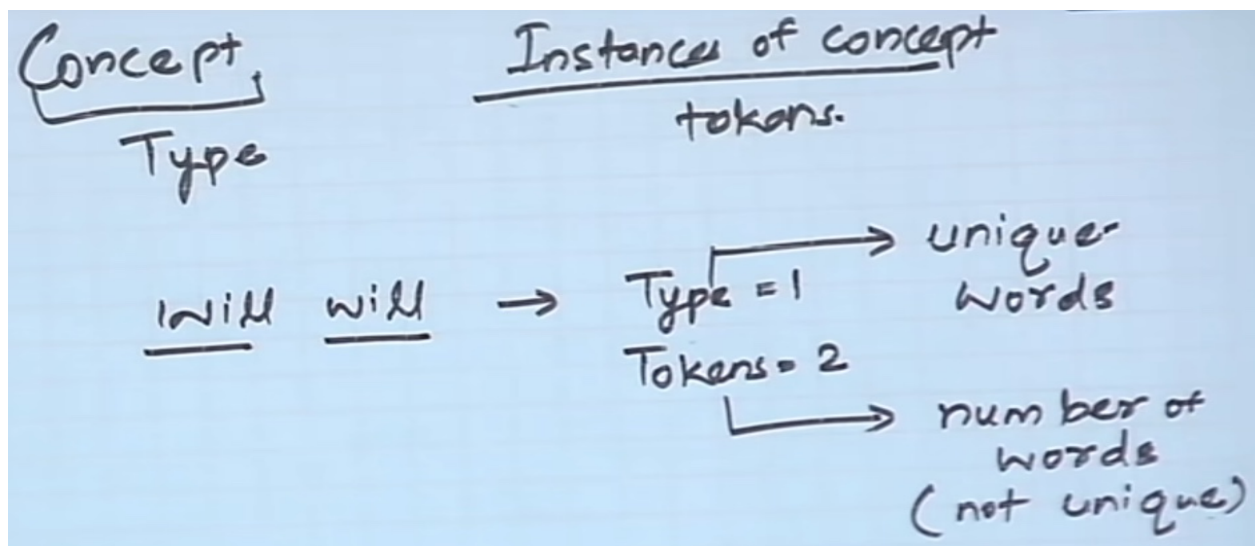
What is one exceptions?

- **Tom is very specific to the topic** (makes to the list of top words)

These are usually referred to as function words, such as determiners, prepositions, complementizers etc.

The one really exceptional word is Tom, whose frequency reflects the text chosen.

How many words are there in this text?



Type vs. Tokens

Type-Token distinction

- Type-token distinction is a distinction that separates a concept from the objects which are particular instances of the concept

Type/Token Ratio

- The type/token ratio (TTR) is the ratio of the number of different words (types) to the number of running words (tokens) in a given text or corpus.
- This index indicates how often, on average, a new 'word form' appears in the text or corpus.

Comparison Across Texts

Mark Twain's Tom Sawyer

- 71,370 word tokens
- 8,018 word types
- TTR = 0.112

Complete Shakespeare work

- 884,647 word tokens
- 29,066 word types
- TTR = 0.032

Empirical Observations on Various Texts

Comparing Conversation, academic prose, news, fiction

- TTR scores the lowest value (tendency to use the same words) in conversation.
- TTR scores the highest value (tendency to use different words) in news.
- Academic prose writing has the second lowest TTR.

Not a valid measure of 'text complexity' by itself

- The value varies with the size of the text.
- For a valid measure, a running average is computed on consecutive 1000-word chunks of the text.

w1L5: Text Processing Basics

Text processing: tokenization

What is Tokenization?

Tokenization is the process of segmenting a string of characters into words.

Depending on the application in hand, you might have to perform sentence segmentation as well.

Sentence Segmentation

The problem of deciding where the sentences begin and end.

Challenges Involved

- While '!', '?' are quite unambiguous
- Period "." is quite ambiguous and can be used additionally for
 - Abbreviations (Dr., Mr., m.p.h.)
 - Numbers (2.4%, 4.3)

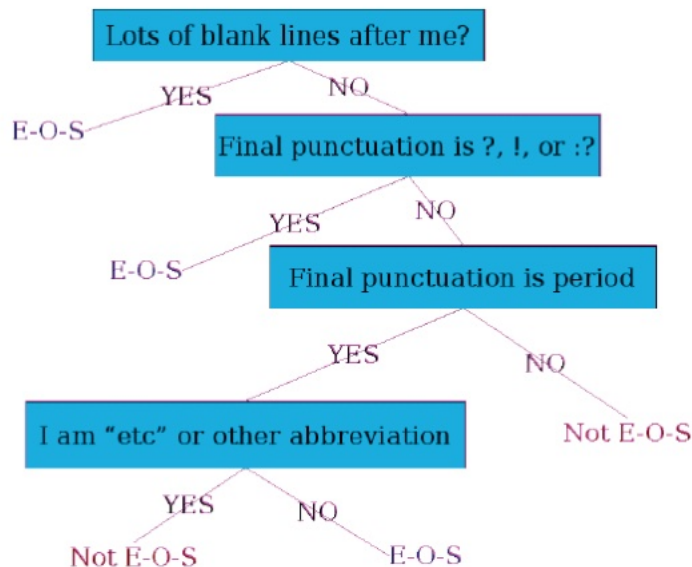
Approach: build a binary classifier

For each "."

- Decides EndOfSentence/NotEndOfSentence
- Classifiers can be: hand-written rules, regular expressions, or machine learning

Sentence Segmentation: Decision Tree Example

Decision Tree: Is this word the end-of-sentence (E-O-S)?



Other Important Features

- Case of word with ".": Upper, Lower, Cap, Number 4.3 D8.
- Case of word after ".": Upper, Lower, Cap, Number
- Numeric Features
 - Length of word with "."
 - Probability (word with "." occurs at end-of-sentence)
 - Probability (word after "." occurs at beginning-of-sentence)

Implementing Decision Trees

- Just an if-then-else statement
- Choosing the features is more important
- For numeric features, thresholds are to be picked
- With increasing features including numerical ones, difficult to set up the structure by hand
- Decision Tree structure can be learned using machine learning over a training corpus

Basic Idea

Usually works top-down, by choosing a variable at each step that best splits the set of items.

Popular algorithms: ID3, C4.5, CART

Other Classifiers

The questions in the decision tree can be thought of as features, that could be exploited by any other classifier:

- Support Vector Machines
- Logistic regression
- Neural Networks

Word Tokenization

What is Tokenization?

Tokenization is the process of segmenting a string of characters into words.

I have a can opener; but I can't open these cans.

Word Token

- An occurrence of a word
- For the above sentence, 11 word tokens.

Word Type

- A different realization of a word
- For the above sentence, 10 word types.

Tokenization in practice

- NLTK Toolkit (Python)
- Stanford CoreNLP (Java)
- Unix Commands

Word Tokenization

Issues in Tokenization

- Finland's -> Finland Finlands Finland's ?
- What're, I'm, shouldn't -> What are, I am, should not ?
- San Francisco ! one token or two?
- m.p.h. -> ??

For information retrieval, use the same convention for documents and queries

Handling Hyphenation

Hyphens can be

End-of-Line Hyphen

Used for splitting whole words into part for text justification.

This paper describes MIMIC, an adaptive mixed initia-tive spoken dialogue system that provides movie show-time information.

Lexical Hyphen

Certain prefixes are often written hyphenated, e.g. co-, pre-, meta-, multi-, etc.

Sententially Determined Hyphenation

- Mainly to prevent incorrect parsing of the phrase. Some possible usages:
- Noun modified by an 'ed'-verb: *case-based, hand-delivered*
- Entire expression as a modifier in a noun group: *three-to-five-year direct marketing plan*

Language Specific Issues: French and German

French

l'ensemble: want to match with un ensemble

German

Noun compounds are not segmented

- Lebensversicherungsgesellschaftsangestellter
- 'life insurance company employee'
- Compound splitter required for German information retrieval

Language Specific Issues: Chinese and Japanese

No space between words

莎拉波娃现在居住在美国东南部的佛罗里达。

莎拉波娃 现在 居住 在 美国 东南部 的 佛罗里达

Sharapova now lives in US southeastern Florida

Japanese: further complications with multiple alphabets intermingled.

フォーチュン500社は情報不足のため時間あた\$500K(約6,000万円)

Katakana Hiragana Kanji Romaji

In chinese, japanese - you just concatenate words

Language Specific Issues: Sanskrit

सत्यम्ब्रूयात्प्रियम्ब्रूयान्ब्रूयात्सत्यमप्रियम्प्रियञ्चनानृतम्ब्रूयादेषधर्मःसनातनः

satyaṃbrūyātpriyaṃbrūyānnabrūyātsatyamapriyaṃpriyaṃcanāṇṛtambrūyād-
eṣadharmahsanātanaḥ.

In Sanskrit: sandhi is done, not just concatenation!

“One should tell the truth, one should say kind words; one should neither tell harsh truths, nor flattering lies; this is a rule for all times.”

Segmented Text:

satyaṃ brūyāt priyaṃ brūyāt na brūyāt satyaṃ apriyaṃ priyaṃ ca na anṛtam
brūyāt eṣaḥ dharmah sanātanaḥ.

Longest Words

Max ▾	Language (non scientific) ⇅		
431	Sanskrit (Longest)	44	Romanian
173	Greek	42	Georgian
136	Afrikaans	41	Czech
85	Māori	39	Bulgarian
79	German	39	Lithuanian
74	Turkish	36	Kazakh
64	Icelandic	33	Norwegian
56	Hungarian	32	Tagalog
54	Spanish	32	Polish
49	Dutch	30	Serbian
46	Malay	30	Montenegrin
45	English	30	Italian
		30	Croatian

Compound word composed of 431 letters, from the Varadāmbikā Parīṇaya Campū by Tirumalāmba

निरन्तरान्धकारिता-दिगन्तर-कन्दलदमन्द-सुधारस-बिन्दु-सान्द्रतर-घनाघन-वृन्द-सन्देहकर-
स्यन्दमान-मकरन्द-बिन्दु-बन्धुरतर-माकन्द-तरु-कुल-तल्प-कल्प-मृदुल-सिकता-जाल-जटिल-
मूल-तल-मरुवक-मिलदलघु-लघु-लय-कलित-रमणीय-पानीय-शालिका-बालिका-करार-विन्द-
गलन्तिका-गलदेला-लवङ्ग-पाटल-घनसार-कस्तूरिकातिसौरभ-मेदुर-लघुतर-मधुर-शीतलतर-
सलिलधारा-निराकरिष्णु-तदीय-विमल-विलोचन-मयूख-रेखापसारित-पिपासायास-पथिक-
लोकान्

Word Tokenization in Chinese or Sanskrit

Also called 'Word Segmentation'.

Greedy Algorithm for Chinese

Maximum Matching (Greedy Algorithm)

- Start a pointer at the beginning of the string
- Find the largest word in dictionary that matches the string starting at pointer
- Move the pointer over the word in string

Think of the cases when word segmentation would be required for English Text?

Finding constituent words in a **compound hashtags**:

#ThankYouSachin, #musicmonday etc.

General assumption behind the design

- W: vocabulary of (inflected) words (*padas*) and
- R: sandhi

A candidate sentence w is analyzed by inverting relation R to produce a finite sequence $w_1, w_2, \dots, w_n$ of word forms, together with a proof that $w \in R(w_1 \cdot w_2 \dots \cdot w_n)$.

Sentence: सत्यम्ब्रूयात्प्रियम्ब्रूयान्नब्रूयात्सत्यमप्रियम्प्रियश्चनानृतम्ब्रूयादेशधर्मःसनातनः

✓Undo (120 Solutions)

satyam brūyāt priyam brūyāt na brūyāt satyam a priyam priyam cana itam brūyāt eṣa dharmas san ā tanah
✓ ✓ ✓ ✓X ✓ ✓ ✓X ✓X ✓ ✓ ✓X ✓ ✓ ✓X
brūyām satī ama ca na an
✓X ✓X ✓X ✓X ✓X
san ātanah
✓X ✓X ✓X ✓X ✓X
sa na
āta
a

Why to “normalize”?

- U.S.A. and USA should be matched
- We implicitly define equivalence classes of terms

- Reduce all letters to lower case
- Possible exceptions (Task dependent):
 - Upper case in mid sentence, may point to named entities (e.g. General Motors)
 - For MT and information extraction, some cases might be helpful (US vs. us)

2. Lemmatization

- Reduce inflections or variant forms to base form:
 - am, are, is -> be
 - car, cars, car's, cars' -> car
- Have to find the correct dictionary headword form

3. Morphology

Morphology studies the internal structure of words, how words are built up from smaller meaningful units called **morphemes**

Morphemes are divided into two categories

- Stems: The core meaning bearing units
- Affixes: Bits and pieces adhering to stems to change their meanings and grammatical functions
 - Prefix: un-, anti-, etc (a-, ati-, pra- etc.)
 - Suffix: -ity, -ation, etc (-taa, -ke, -ka etc.)
 - Infix: 'n' in 'vindati' (he knows), as contrasted with *vid* (to know).

4. Stemming

- Reducing terms to their stems, used in information retrieval
- Crude chopping of affixes
 - language dependent
 - automate(s), automatic, automation all reduced to automat

*for example compressed
and compression are both
accepted as equivalent to
compress.*



for exampl compress and
compress ar both accept
as equival to compress

Porter's algorithm

Step 1a

- sses → ss (caresses → caress)
- ies → i (ponies → poni)
- ss → ss (caress → caress)
- s → ϕ (cats → cat)

Step 1b

- (*v*)ing → ϕ (walking → walk, king → king)
- (*v*)ed → ϕ (played → play)

Step 2

- ational → ate (relational → relate)
- izer → ize (digitizer → digitize)
- ator → ate (operator → operate)

Step 3

- al → ϕ (revival → reviv)
- able → ϕ (adjustable → adjust)
- ate → ϕ (activate → activ)